Write your name here

Surname	Other names
Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	FSM01/01

Mathematics

Level 1



9 – 13 January 2017
Time: 1 hour 30 minutes

Paper Reference
FSM01/01

You must have:
Pen, calculator, HB pencil, eraser, ruler graduated in cm and mm, protractor, compasses.

Total Marks
48 /48

My signature confirms that I will not discuss the content of the test with anyone until the end of the 5 day test window.

Signature: Master Functional Skills (MFS)

Instructions

- Use a **black** ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Answer the questions in the spaces provided
  – there may be more space than you need.
- **Calculators may be used.**

Information

- The total mark for this paper is 48.
- The marks for each question are shown in brackets
  – use this as a guide as to how much time to spend on each question.
- **You must show clearly how you get your answers because marks will be awarded for your working out.**
- **Check your working and your answers at each stage.**
- **This sign shows where marks will be awarded for showing your check.**

Advice

- Read each question carefully before you start to answer it.
- Keep an eye on the time.

Turn over ▶

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SECTION A: Birdwatching

Answer all questions in this section.

Write your answers in the spaces provided.

1 Liz wants to feed the birds in her garden.
She wants to buy 2 bird feeders and 3 bags of bird seed.

Each bird feeder costs £6.95
A bag of bird seed costs £3.10

(a) Work out the total cost for Liz.

(3)

Use the box below to show clearly how you get your answer.

Step 1: Calculate cost of bird feeders:

- *2 bird feeders at £6.95 each*
- **Cost = 2 × £6.95 = £13.90 [M1]**

Step 2: Calculate cost of bird seed:

- *3 bags of bird seed at £3.10 each*
- **Cost = 3 × £3.10 = £9.30 [M1]**

Step 3: Calculate total cost for Liz:

- **Total = £13.90 + £9.30 = £23.20 [M1]**

Final Answer:

Total cost = £23.20 [A1]

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Liz and her family want to join a bird club for a year.

Bird Club Membership

Adult £4 per month
Child £3 per month

Bird Club Family Membership

1 or 2 adults + up to 3 children
£126 per year

Liz and her husband are adults.
Andy is their only child.
They want to choose the cheapest membership deal for their family.

(b) Which membership deal should they choose?
Show a check of your working. **(4)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate monthly cost of individual memberships (2 adults, 1 child):

- *2 adults = 2 × £4 = £8 per month*
- *1 child = £3 per month*
- ***Total per month = £8 + £3 = £11 per month [M1]***

Step 2: Calculate annual cost for individual memberships:

- ***12 months in a year: 12 × £11 = £132.00 for the year [M1]***

Step 3: Compare with Family membership deal (£126 per year):

- *Family membership = £126 per year*
- ***£126 < £132 (Family membership saves £132 − £126 = £6 per year). [M1]***

(Or monthly comparison: £126 ÷ 12 = £10.50/month < £11.00/month).

Final Decision: They should choose the Family membership. [A1]

Write your check in the box below.

☑

Reverse Process & Rate Check: [B1]

- *Reverse multiplication: £132 ÷ 12 = £11 per month.*
- *Subtract child: £11 − £3 = £8 for 2 adults; £8 ÷ 2 = £4 per adult.*
- *Annual savings check: £132 − £126 = £6 cheaper with Family deal.*

(Total for Question 1 is 7 marks)

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2 Matt is building a bird box from wood.

The side of the bird box is shown below.

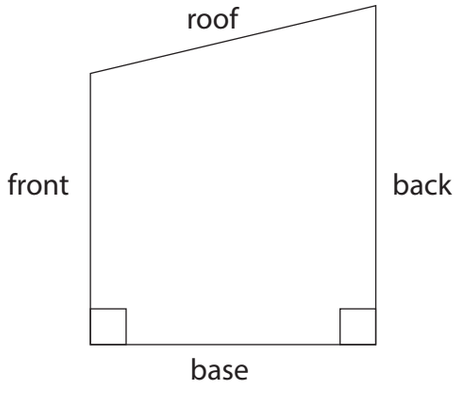


Diagram **not** accurately drawn

The length of the base is 120 mm.
The height of the front is 150 mm.
The height of the back is 200 mm.

(a) Make an accurate drawing of the side of the bird box for Matt.

(3)

Use the space opposite for your drawing.
The base has been drawn for you.

(b) Measure the length of the roof.
Give the units with your answer.

(1)

Write your answer in the box below.

Measure Length of Roof from Scale Drawing on Opposite Page: [B1]

- *Measured length = 13 cm (or 130 mm) [B1]*
- *Geometric Calculation Check (Pythagoras):*

  Horizontal distance = 120 mm; Vertical difference = 200 − 150 = 50 mm

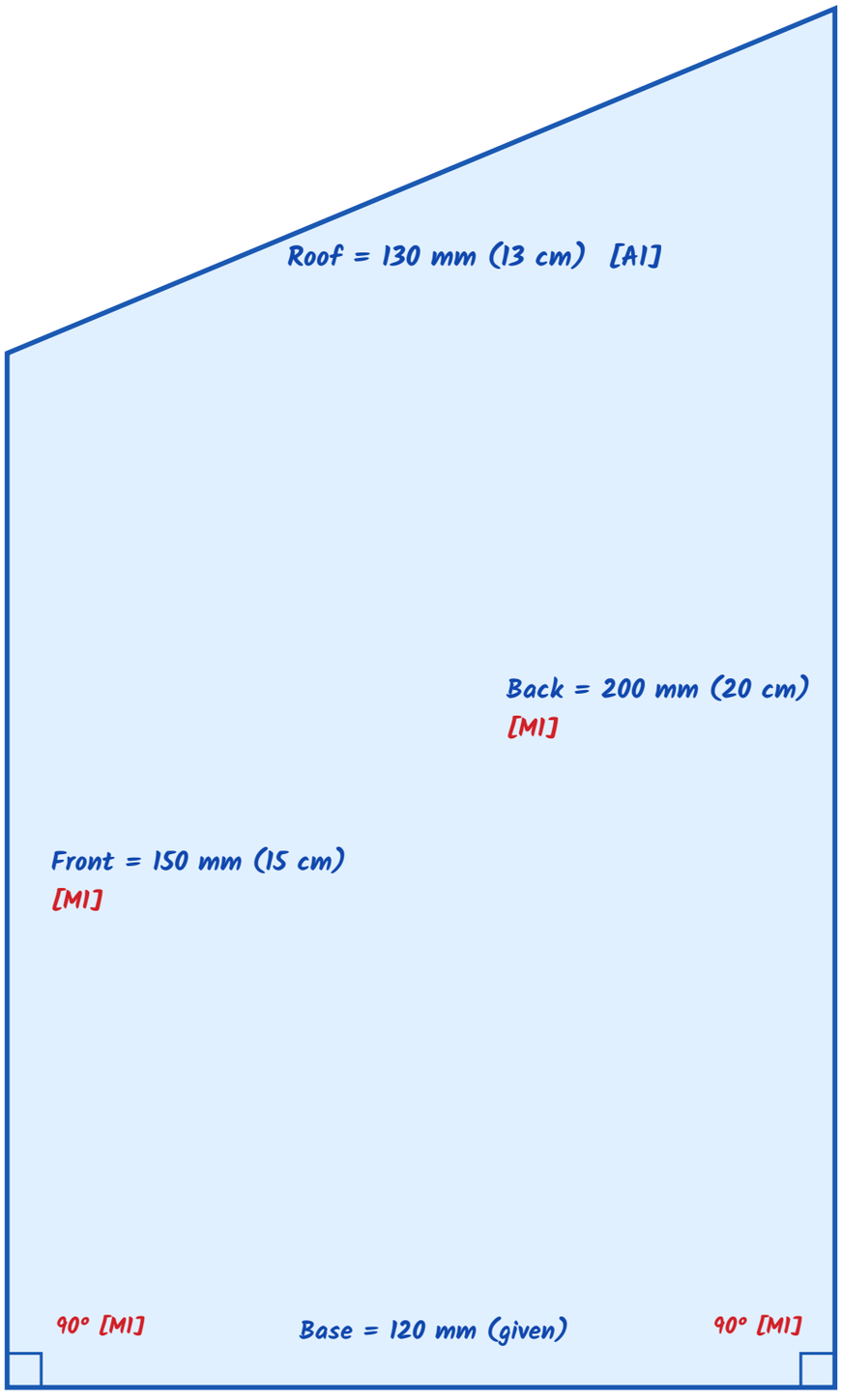
  Roof = √(120² + 50²) = √(14400 + 2500) = √16900 = 130 mm = 13 cm

Final Answer: 13 cm (or 130 mm) [B1]

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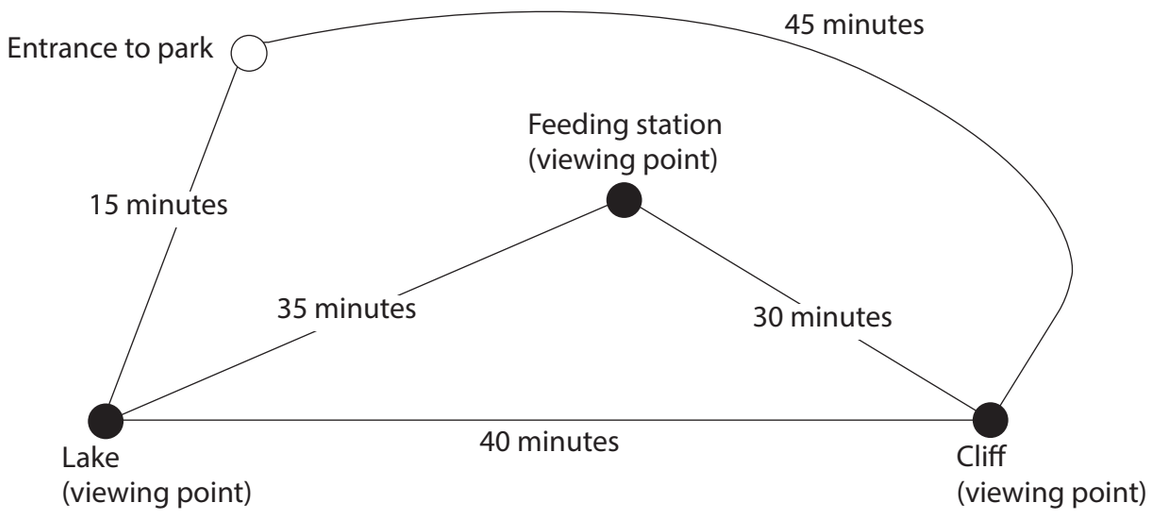
base

(Total for Question 2 is 4 marks)

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3 Liz is going to spend the day birdwatching at a park.

She has this plan of the paths in the park.
The plan shows the time it takes to walk along each path.



Liz arrives at the entrance to the park at 9 am.

She will walk in the park without going back along a path she has already used.

She will spend 1 hour at each of the viewing points.

Liz thinks she will be back at the entrance by 2 pm.

Is Liz correct?
Show why you think this. **(5)**

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Use the box below to show clearly how you get your answer.

Step 1: Calculate total time available between 9 am and 2 pm:

- **9:00 am to 2:00 pm = 5 hours = 300 minutes available [M1]**

Step 2: Calculate total viewing time needed at the 3 hides:

- **3 hides (A, B, and C) × 1 hour each = 3 hours = 180 minutes [M1]**
- Remaining time for walking = 300 − 180 = 120 minutes (2 hours maximum).

Step 3: Calculate walking times for circuit visiting all 3 hides:

- **Circuit Route: Entrance -> Hide A -> Hide C -> Hide B -> Entrance**
  - Entrance to Hide A = 15 minutes [M1]
  - Hide A to Hide C = 35 minutes
  - Hide C to Hide B = 40 minutes
  - Hide B to Entrance = 45 minutes
- **Total walking time = 15 + 35 + 40 + 45 = 135 minutes (2 hr 15 mins) [M1]**
  (135 mins walking > 120 mins available for walking).

Step 4: Full Sequential Itinerary & Timetable:

• 09:00: Arrive at park entrance	**(Start)**
• 09:00 – 09:15: Walk from Entrance to Hide A	**(15 mins walk)**
• 09:15 – 10:15: Spend 1 hour at Hide A	**(60 mins viewing)**
• 10:15 – 10:50: Walk from Hide A to Hide C	**(35 mins walk)**
• 10:50 – 11:50: Spend 1 hour at Hide C	**(60 mins viewing)**
• 11:50 – 12:30: Walk from Hide C to Hide B	**(40 mins walk)**
• 12:30 – 13:30: Spend 1 hour at Hide B	**(60 mins viewing)**
• 13:30 – 14:15: Walk from Hide B to Entrance	**(45 mins walk)**
• 14:15: Arrive back at Entrance (2:15 pm)	**(Total: 5 hr 15 min [M1])**

Final Conclusion & Decision: [A1]

No, Liz cannot visit all 3 bird hides and spend 1 hour at each.
The visit requires 5 hours 15 minutes (finishing at 2:15 pm).
This is 15 minutes past her departure deadline of 2:00 pm. [A1]

(Total for Question 3 is 5 marks)

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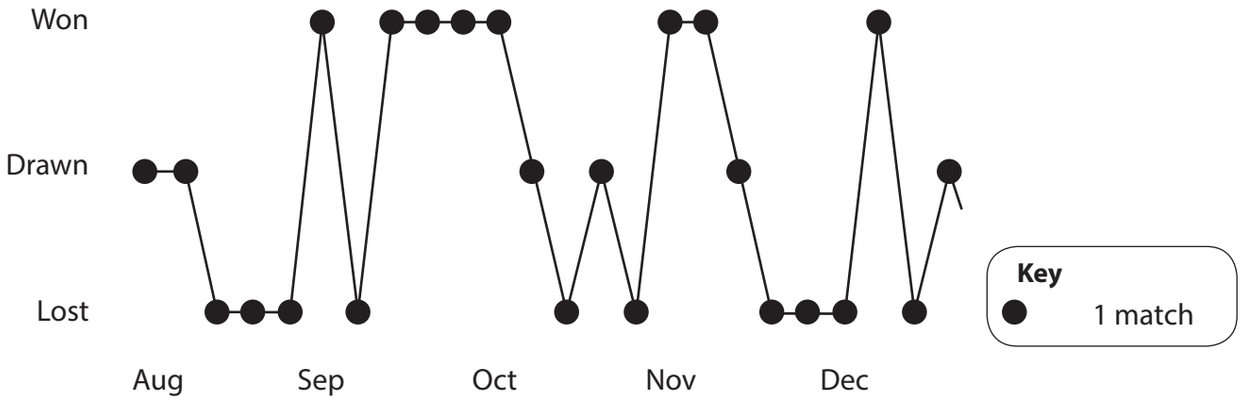
SECTION B: Lowton United Football Club

Answer all questions in this section.

Write your answers in the spaces provided.

4 Lowton United is a football club.

The diagram shows some information about the matches Lowton United played in the league competition from August to December.



A team gets

- 3 points for each match won
- 1 point for each match drawn
- 0 points for each match lost.

(a) How many points in total did Lowton United get from the beginning of August to the end of December? **(3)**

Use the box below to show clearly how you get your answer.

Step 1: Read match outcomes from the chart:

- *Wins = 8; Draws = 6; Losses = 10 (Total matches = 24) [M1]*

Step 2: Calculate points for each outcome:

- ***Points from wins = 8 × 3 = 24 points [M1]***
- ***Points from draws = 6 × 1 = 6 points***
- *Points from losses = 10 × 0 = 0 points*

Final Answer: Total points = 24 + 6 + 0 = 30 points [A1]

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Lowton United is taking part in a cup competition.

There are 8 teams including Lowton United in the competition.

To decide who will play who in the next round, the team names are chosen at random from a bag.

Only one team is chosen at a time.

(b) What is the likelihood that Lowton United is chosen from the bag first?

(1)

Tick ✔ a box to show your answer.

impossible	☐
unlikely	☑ [B1]
even chance	☐
likely	☐
certain	☐

Probability of Lowton United chosen first = 1/8 = 12.5% (Unlikely)

(Total for Question 4 is 4 marks)

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5 Paulo manages the ticket sales for Lowton United.
He is writing a report about ticket sales.

He wants to draw a graph to show the ticket sales for the last 6 matches.

Ticket sales					
match 1	match 2	match 3	match 4	match 5	match 6
6200	5600	6400	6800	7200	8300

Paulo starts to draw a graph to show this information.

(a) Complete the graph for Paulo.

(3)

Use the graph paper and the scale on the opposite page.

Paulo wants to write a comment in the report about the ticket sales.

(b) Write a comment for the report.

(1)

Write your comment in the box below.

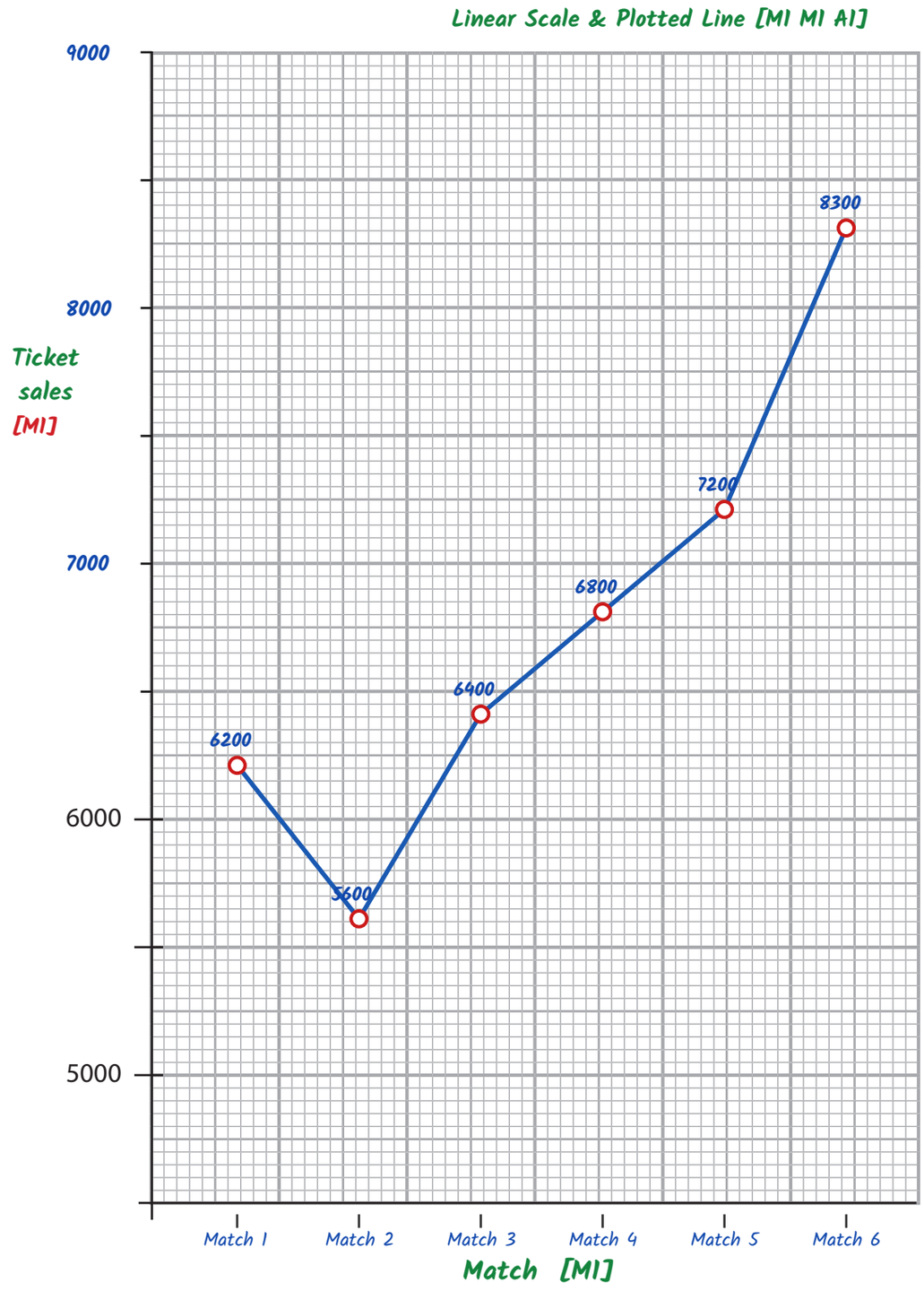
Report Comment on Ticket Sales Trend: [B1]

- *Ticket sales show an overall increasing trend from Match 2 onwards.*
- *Sales steadily rise from 5,600 (Match 2) to a peak of 8,300 in Match 6.*
- ***Match 6 had the highest attendance of the last 6 matches. [B1]***

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(Total for Question 5 is 4 marks)

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6 The total ticket sales for each of the last 6 matches are shown in the table below.

Ticket sales					
match 1	match 2	match 3	match 4	match 5	match 6
6200	5600	6400	6800	7200	8300

Paulo thinks the average ticket sales for these matches is over 6500

(a) Is Paulo correct?
Show a check of your working. **(4)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate total ticket sales for the 6 matches:

- *6400 + 5600 + 6200 + 6800 + 7200 + 8300*
- ***Total sales = 40,500 tickets [M1]***

Step 2: Calculate the mean ticket sales:

- ***Mean = Total sales ÷ 6 = 40,500 ÷ 6 = 6,750 tickets [M1]***

Step 3: Compare mean with 6,500:

- ***6,750 > 6,500 (the mean is 250 tickets above 6,500). [M1]***

(Alternative: 6 × 6,500 = 39,000; 40,500 > 39,000).

Final Conclusion:

Yes, Paulo is correct. The mean is 6,750 tickets (exceeds 6,500). [A1]

Use the box below to show your check.

☑

Reverse Calculation Check: [B1]

- *Reverse division by multiplication: 6,750 × 6 = 40,500 tickets.*
- *Benchmark check: 6 × 6,500 = 39,000; 40,500 − 39,000 = 1,500 extra tickets.*

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The table below shows the prices and the ticket sales for the last match.

	Price	Ticket sales
Adult	£22	4600
Juniors (under 16)	£8	3700

The football club wants the income from ticket sales for the last match and the next match to be the same.

For the next match the club is going to

- give free tickets to juniors
- charge £25 for each adult ticket.

(b) How many adult tickets does the club need to sell to keep the income from ticket sales the same? **(4)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate income from adult tickets:

- *4,600 adult tickets at £22 each*
- *Adult income = 4,600 × £22 = £101,200 [M1]*

Step 2: Calculate income from junior tickets:

- *3,700 junior tickets at £8 each*
- *Junior income = 3,700 × £8 = £29,600 [M1]*

Step 3: Calculate total ticket income from last match:

- *Total income = £101,200 + £29,600 = £130,800 [M1]*

Step 4: Calculate number of £25 tickets needed to maintain income:

- *All tickets next season = £25 each*
- *Tickets needed = £130,800 ÷ 25 = 5,232 tickets [M1]*

Final Answer:

The club needs to sell 5,232 tickets. [A1]

(Total for Question 6 is 8 marks)

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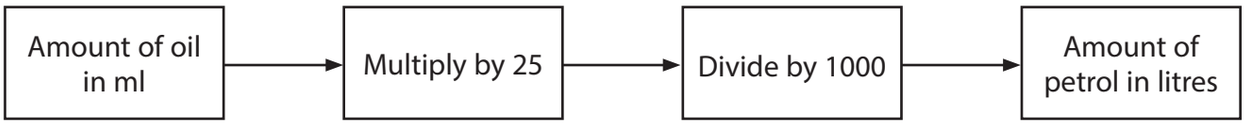
SECTION C: Lawn care

Answer all questions in this section.

Write your answers in the spaces provided.

7 Jane needs to mix petrol and oil for her lawn mower.

She finds these instructions for mixing the petrol and oil.



Jane uses 100 ml of oil.

(a) How much petrol should Jane use?
Show a check of your working. **(3)**

Use the box below to show clearly how you get your answer.

Step 1: Follow the instruction rule with 100 ml of oil:

- *Rule: Amount of oil (ml) × 25 ÷ 1000 = Petrol (litres)*

Step 2: Calculate amount of petrol:

- *Step 2a: 100 × 25 = 2,500 [M1]*
- *Step 2b: 2,500 ÷ 1,000 = 2.5 litres [M1]*

Final Answer:

Jane should use 2.5 litres of petrol. [A1]

Use the box below to show your check.

☑

Reverse Process Calculation Check: [B1]

- *Reverse division by multiplication: 2.5 litres × 1,000 = 2,500 ml.*
- *Reverse multiplication by division: 2,500 ÷ 25 = 100 ml of oil.*
- *Matches the original 100 ml of oil used.*

✓

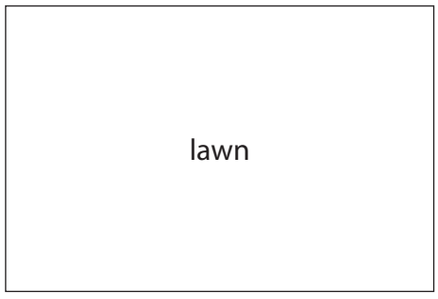
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Jane has a lawn in front of her house.

The lawn is in the shape of a rectangle 6 m by 4 m.



Jane buys one box of lawn feed.
One box is enough for 100 m^2 of lawn.

Jane says

My lawn needs $\frac{1}{4}$ of this box.

(b) Will $\frac{1}{4}$ of this box be enough for the lawn?

(4)

Use the box below to show clearly how you get your answer.

Step 1: Calculate the area of the rectangular lawn:

- Dimensions: length = 6 m, width = 4 m
- **Area of lawn = 6 m × 4 m = 24 m^2 [M1]**

Step 2: Calculate coverage of 1/4 of a box of lawn feed:

- 1 full box covers 100 m^2
- **1/4 of a box = 100 m^2 ÷ 4 = 25 m^2 [M1]**

Step 3: Compare lawn area with coverage of 1/4 box:

- **25 m^2 > 24 m^2 (1/4 box covers 1 m^2 more than the lawn). [M1]**

(Alternative: 24 ÷ 100 = 0.24 = 24%; 24% < 25%).

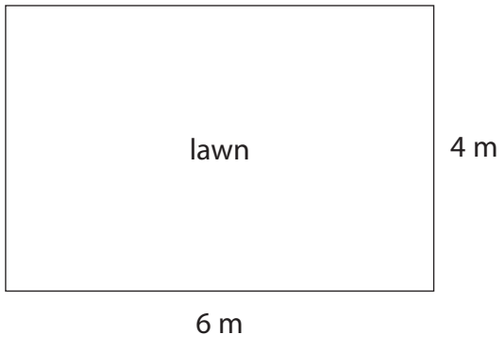
Final Conclusion:

Yes, 1/4 of this box will be enough (covers 25 m^2 vs 24 m^2 needed). [A1]

(Total for Question 7 is 7 marks)

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8 Jane wants to put edging around her lawn.



The edging is sold in 50 cm strips.
The edging strips cost £3 each.

Jane wants to know how much it will cost to put edging strips around the lawn.

Find the total cost of the edging strips for Jane.

(5)

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Use the box below to show clearly how you get your answer.

Step 1: Calculate total perimeter of the rectangular lawn:

- Dimensions: length = 6 m, width = 4 m
- **Perimeter = 6 + 4 + 6 + 4 = 20 metres [M1]**

Step 2: Convert units to find the number of edging strips needed:

- Each edging strip is 50 cm = 0.5 metres [M1]
- **Number of strips = 20 m ÷ 0.5 m = 40 strips [M1]**

(Alternative side-by-side check: along two 6 m sides = 2 × (6 ÷ 0.5) = 24 strips; along two 4 m sides = 2 × (4 ÷ 0.5) = 16 strips; Total = 24 + 16 = 40 strips).

Step 3: Calculate the total cost of 40 edging strips:

- Each strip costs £3
- **Total cost = 40 strips × £3 = £120.00 [M1]**

Final Answer:

Total cost of edging strips = £120.00 [A1]

(Total for Question 8 is 5 marks)

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9 Jane wants to buy a barbecue.

She sees this offer.

Barbecue £150
Pay 20% deposit.

(a) Work out 20% of £150 for Jane.

(2)

Use the box below to show clearly how you get your answer.

Step 1: Calculate 20% deposit on £150:

- *20% = 20 ÷ 100 = 0.20 (or 1/5) [M1]*
- ***Deposit = 0.20 × £150 = £30.00***

(Alternative: 10% of £150 = £15; 20% = 2 × £15 = £30.00).

Final Answer: 20% deposit = £30.00 [A1]

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Jane also wants to buy some garden furniture.

The garden furniture costs £800

She can pay for the garden furniture in 6 equal monthly payments.

Jane thinks she will need to pay £140 each month.

(b) Is Jane correct?
Show why you think this. **(2)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate total paid with 6 payments of £140:

- Monthly payment = £140
- **6 equal monthly payments = 6 × £140 = £840.00 [M1]**

Step 2: Compare total paid with actual furniture cost (£800):

- Actual cost = £800.00
- **£840.00 > £800.00 (Jane would overpay by £840 − £800 = £40.00).**
- Alternative calculation: actual monthly payment = £800 ÷ 6 = £133.33...
  £133.33 < £140.00 per month.

Final Conclusion & Decision:

No, Jane is not correct.

6 payments of £140 equal £840, which is £40 more than the £800 cost.

(She only needs to pay £133.33 per month, not £140). [A1]

(Total for Question 9 is 4 marks)

TOTAL FOR PAPER IS 48 MARKS

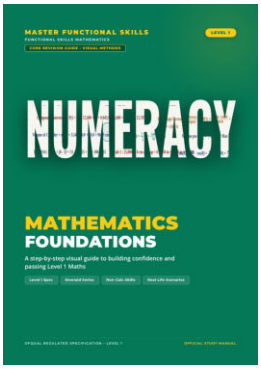
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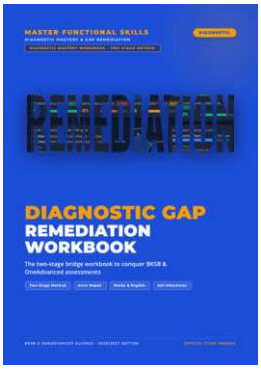


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